Name	Jenica A	Register No.	192425003
Course	ITA0510 - COMPUTER VISION	Experiment No.	20
Programme	B.Tech. Artificial Intelligence and Machine Learning - III Year	Date	15-07-2026
Institution	Saveetha School of Engineering, SIMATS	Signature	

EXPERIMENT 20

Sharpening an Image using a Laplacian Mask with a Negative Center Coefficient

AIM

To sharpen the given image using a Laplacian mask whose center coefficient is negative.

PROGRAM

```
import cv2
import numpy as np

img = cv2.imread("image1.jpg")

# 4-neighbour Laplacian, negative center
kernel = np.array([[0,  1, 0],
                   [1, -4, 1],
                   [0,  1, 0]], dtype=np.float32)

laplacian = cv2.filter2D(img, cv2.CV_64F, kernel)

# The center coefficient is negative, so the mask is SUBTRACTED: g = f - Lap(f)
sharpened = np.clip(img - laplacian, 0, 255).astype(np.uint8)

cv2.imwrite("laplacian_negative_center.jpg", sharpened)
cv2.imshow("Original", img)
cv2.imshow("Sharpened", sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT

image1.jpg (input image)

OUTPUT

laplacian_negative_center.jpg (sharpened image, g = f - Laplacian(f))

RESULT

Thus the given image was successfully sharpened using a Laplacian mask with a negative center coefficient.